Green Patent Detection: Advanced Agentic Workflow with QLoRA

Using PatentSBERTa, QLoRA Fine-Tuning, and Multi-Agent Debate

Applied Deep Learning and Artificial Intelligence

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Roadmap

1 Neural Network

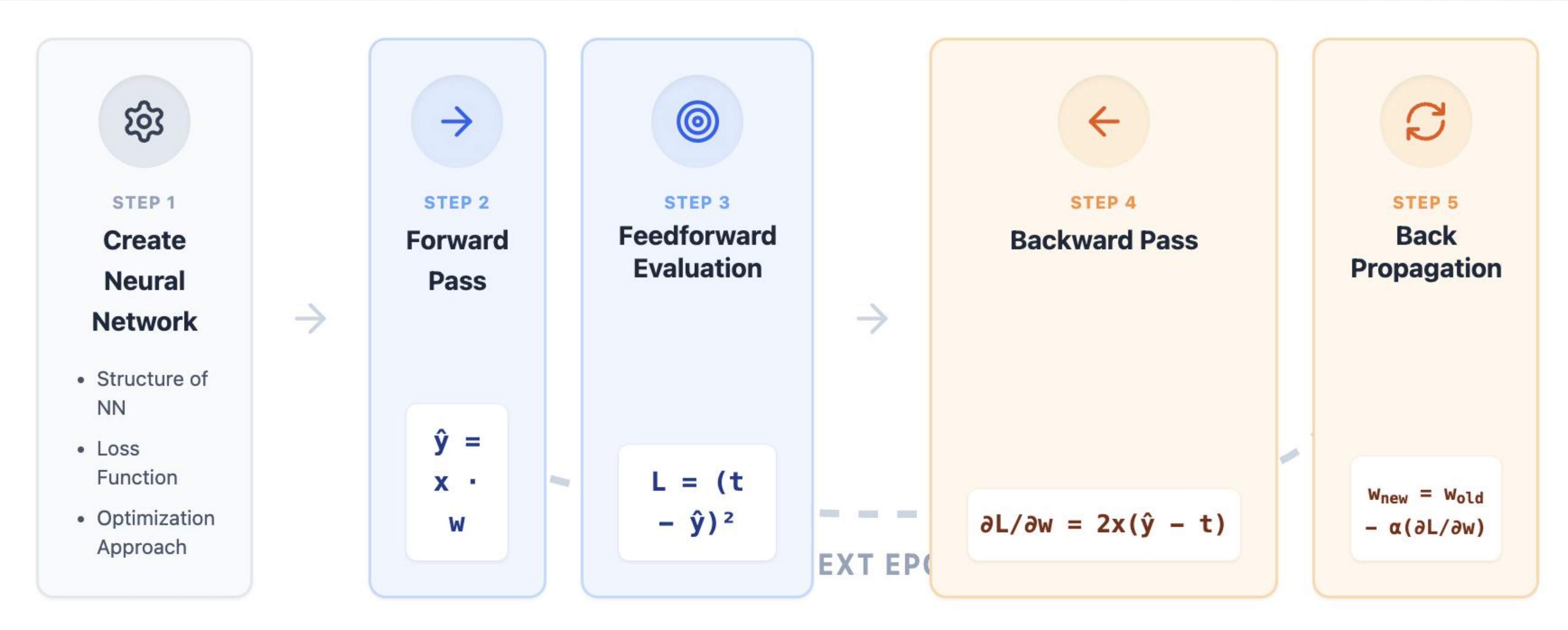
2 Self-Attention

3 Active Learning

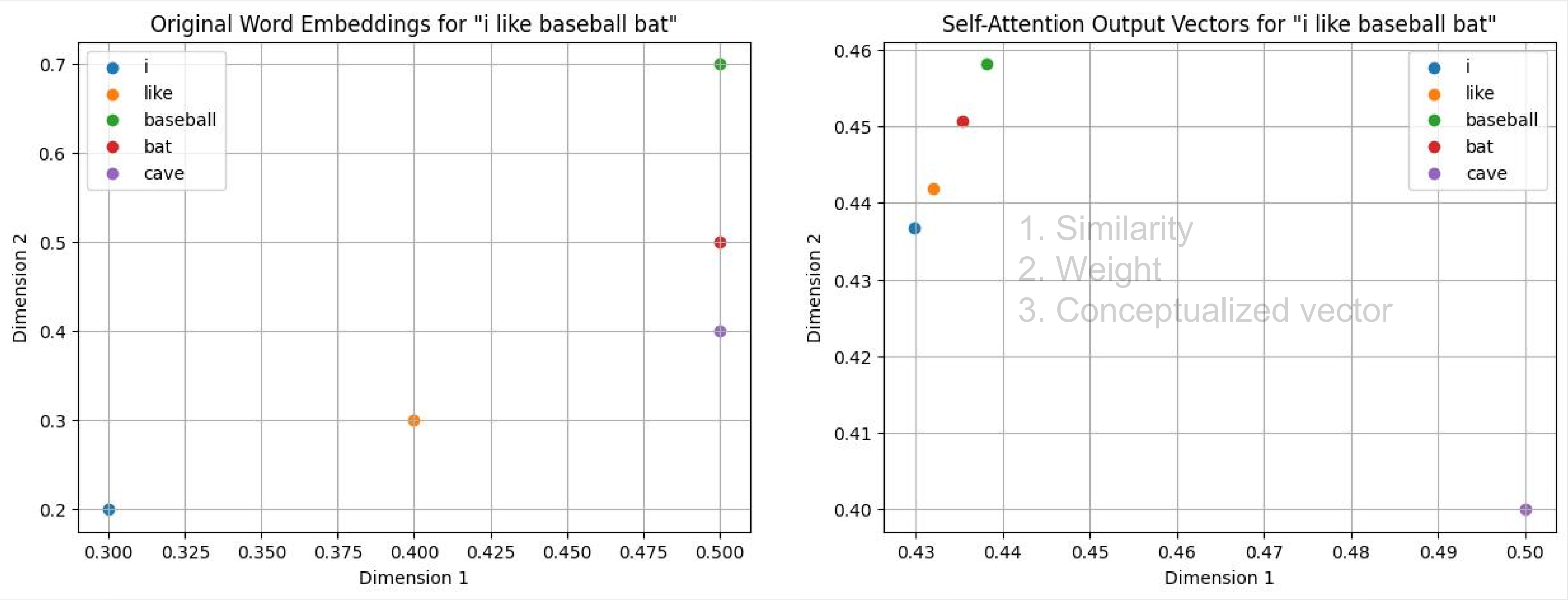
4 Advanced Architecture

5 Finetune

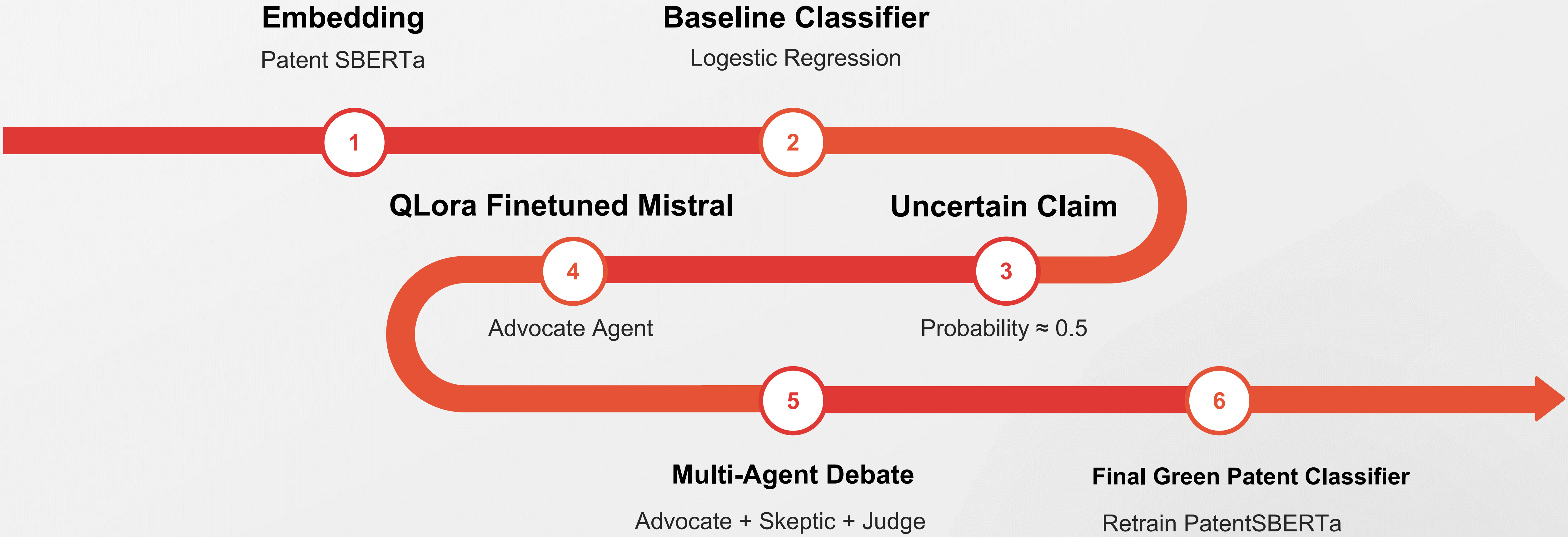
Neural Network



Self-Attention



Overview


Embedding
Patent SBERTa
1
Baseline Classifier
Logestic Regression
2
Uncertain Claim
3
Probability ≈ 0.5
QLora Finetuned Mistral
4
Advocate Agent
5
Multi-Agent Debate
Advocate + Skeptic + Judge
6
Final Green Patent Classifier
Retrain PatentSBERTa

Active Learning

Dataset Overview

Total_Dataset: 1.5M

Sample_size: 50k

train_silver: 35k

eval_silver:7.5k

pool_unlabeled: 7.5k

Baseline model

Patent Text

↓

Sentence Embeddings

(PatentSBERTa)

↓

Logistic Regression

↓

Prediction

Uncertainty Sampling

p_green

↓

compute u = 1 - 2|p-0.5|

↓

select top 100 uncertain

patents

Advanced Architecture

Assignment 1

Uncertain Claim

↓

Ollama LLM (Qwen 2.5)

↓

Structured JSON Output

Assignment 2

Uncertain Claim

↓

Advocate Agent

↓

Skeptic Agent

↓

Judge Agent

Assignment 3 (A)

Train Silver

↓

Mistral-7B

↓

LoRA Adapters

↓

Fine-tuned Model

Advanced Architecture

Assignment 1

Uncertain Claim

↓

Ollama LLM (Qwen 2.5)

↓

Structured JSON Output

Assignment 2

Uncertain Claim

↓

Advocate Agent

↓

Skeptic Agent

↓

Judge Agent

Assignment 3 (B)

Uncertain Claim

↓

Advocate Agent
Qlora Finetuned

↓

Skeptic Agent

↓

Judge Agent

Human Review & Final Integration

HITL

Confidence < 0.85

↓

Human Expert Review

↓

Correct Label

Gold dataset = LLM predictions + Human corrections

Final Integration

Silver Labels

+

Human-corrected Gold Labels

↓

Final Training Dataset

↓

PatentSBERTa Fine-Tuning

↓

Final Green Patent Classifier

Model Evaluation

Model Version	Training Data Source	F1 Score
Baseline	Frozen Embeddings (No Fine-tuning)	0.7752
Assignment 2	Fine-tuned on Silver + Gold (Simple Generic LLM)	0.8006
Assignment 3	Fine-tuned on Silver + Gold (Advanced Techniques)	0.8116
Final Model	Fine-tuned on Silver + Gold (QLoRA-Powered MAS + Targeted HITL)	0.8053

THANK YOU